Findings: Country Differences

> Part of the Functional Series. Global totals hide very different national stories. This document summarises what the platform's data show for the ten featured countries. The comparisons use gross emissions excluding land use, so they indicate direction and distance, not whether a pledge will be met; the scope limits are set out in Country NDCs §3.
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> **As of:** platform data to 2024 (PRIMAP-hist total GHG, whose country values are rounded to three significant figures; OWID CO_2); UNEP *Emissions Gap Report 2025* for pledges and assessments.

Finding 10. Emissions have peaked in some featured countries and are still rising in others

Change in total greenhouse gas emissions (excluding land use) from **2019** to 2024, the base year of the IPCC pathways, with the year of the series' peak:

Country	Peak of the series	2024 vs 2019	Group
Germany	1979	about −19%	Peaked long ago; falling
United Kingdom	1979	about −16%	Peaked long ago; falling
Japan	around 2007–2013	about −12%	Peaked; falling
Australia	2018	about −7%	Recently peaked; falling slowly
United States	2007	about −6%	Peaked; falling slowly
Russia	1988	about +4%	Well below its peak; rising again since 2019
South Africa	2009	about +4%	Near a plateau
Brazil	2023	about +8%	At or near a record
China	2024	about +13%	At a record high
India	2024	about +16%	At a record high

Evidence. Platform series, Country NDCs §3; Data Sources.

Confidence. High for direction (large differences against rounding of three significant figures). The peak years for Japan, Brazil and the United States are close ties and should not be read to the year.

Does not say. That the falling countries are on track for their targets (Finding 11), or that the rising ones are doing worse *per person* or *per unit of output*. Emissions per capita and per unit of GDP are separate questions the platform shows on other pages.

Finding 11. Falling emissions are not the same as meeting a target

Finding. Three featured countries are measured against 1990 (Germany about –48%, the UK about –52%, Russia about –33%, on gross figures), and they are in different positions against their targets:

- **Russia** is already past its 2030 pledge on a gross basis, and UNEP projects it about 25% below its target level.
- **The UK** is at –52% against a pledge of at least –68% by 2030, and UNEP assesses it as less likely to meet it on current policies (official UK projections agree).
- **Germany** shares the EU's collective NDC (–55% by 2030); on the platform's BAU it reaches about –52% by 2030.
- **Japan** (about –25% against 2013, a pledge of –46%) and **Australia** (a small gross change against a pledge of –43%) are described by UNEP as "very close" to meeting their 2030 targets on current policies. Gross figures can overstate the distance where the pledges are net of land removals.

Evidence. Country NDCs §2–3.

Confidence. Moderate: the direction of each country's gap is clear; the size depends on scope (net versus gross, base year).

Does not say. Anything about which countries will meet their targets. UNEP's policy-based assessment is the right source for that.

Finding 12. Growth is concentrated in China and India, and the trend model is least reliable for China

Finding. China and India show the largest growth since 2019 (about +13% and +16% in total greenhouse gases; China's roughly 1.8 $GtCO_2e$ is the largest in absolute terms) and both are at record highs in 2024. China is also where the platform's statistical forecast most plausibly **overstates** future emissions: its BAU rises about 10% by 2030, whereas UNEP's policy-based studies show a peak around 2025 followed by a fall of 0.3–1.4 $GtCO_2e$ by 2030, driven by renewable generation growing faster than power demand.

Evidence. Country NDCs §2–3; Forecasting §4.

Confidence. Moderate. The statistical model is a continuation of 1990–2024 behaviour and, by design, cannot see policy or technology change. For a country in structural transition, that is the largest single limit on the platform's forecasts.

Does not say. That China's emissions have peaked. In the platform's data, 2024 is the highest year yet.

Finding 13. Pledges measure different things, so they cannot be ranked on one scale

Finding. The ten featured countries' pledges take different forms: China's combines a peak before 2030 with a carbon-intensity target, India's is an emissions-intensity target, South Africa's is an absolute range, and the rest

are percentage reductions from base years running from 1990 (the EU, UK and Russia) to 2013 (Japan). Several are stated net of land removals (Brazil and the EU, for example).

Evidence. Country NDCs §1.

Confidence. High for the description; the consequences are summarised in Caveats.

Does not say. That any pledge is more or less ambitious than another. That needs a common metric, such as per-capita emissions or an equity-based fair share, which the platform does not compute.

See also

- Where Things Stand
- Trajectory Against Paris
- Caveats and Reading Guide